

From the Club of Rome to Star Wars: The Era of Limits, Space Colonization and the Origins of SDI

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In 1972 the Club of Rome released a report titled *The Limits to Growth*. The report heralded what would be known as the ‘era of limits,’ when demographic and environmental pressures would force humanity to scale back its ambitions and live with less.¹ A decade later, in 1983, US President Ronald Reagan announced his plan to build a shield in outer space against ballistic missiles. What became known as the Strategic Defense Initiative (SDI) represented a remarkably ambitious vision of literally high technology erasing the threat of nuclear weapons.

The Club of Rome and SDI are familiar historical events, but they seem entirely disconnected. This chapter connects them, showing how the cultural and intellectual context of the 1970s – specifically, the idea of an impending crisis in human society – helped lead to the military program known as SDI. The path ran through space-colonization groups such as the L5 Society and the science-fiction community. The perceived threat to human survival galvanized a group of technological optimists to propose outer space not only as a solution to the Malthusian crisis of overpopulation, energy depletion and environmental damage, but also to the possibility of nuclear Armageddon. This quest for a technological fix led to surprising political alignments, with countercultural types embracing technology amid the backlash against science and technology

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after the 1960s and Vietnam, and also linking up with libertarians and others on the political right pushing missile defense as well as space colonization. The convergence included interesting individual, institutional and intellectual connections, as space-colony enthusiasts mingled with nuclear-weapon designers, and pro-missile defense arguments cohabited with enlightened discussion of biofeedback and the Gaia hypothesis.

These unusual intersections highlight a historiographical need to look off the beaten path for the origins of SDI. The need is especially acute for SDI, since the path has been trodden heavily by actors from the Reagan Administration in their memoirs and interviews. The current historical literature relies a great deal on these post-facto accounts, which impose an overly coherent perspective focused on high policy and almost exclusively on the United States.² Looking beyond the United States – and outside high government offices – yields new perspectives. The Soviets, for example, took a different view of SDI, one based on their own history and culture, and one that American scientists and strategists were slow to recognize.³ Current scholarship on SDI also concentrates, with good reason, on the context of the Cold War, especially the renewed sense of confrontation of the late 1970s and early 1980s after the collapse of détente, and the resultant redoubling of the arms race. Europe, China and Japan, however, viewed SDI not as a Cold War, US-Soviet strategic standoff, but rather as a high-tech economic stimulus in a new era of global industrial competition.⁴ This chapter keeps the United States in the foreground, but it broadens the focus further to include developments from earlier in the 1970s and beyond the usual Cold War framework. It thus reveals how the era of limits gave rise to the vast ambitions of SDI.

I Space colonies and space forts

The Club of Rome's manifesto was just one of several books in the late 1960s and early 1970s, including *The Population Bomb* and *The Coming Dark Age*, to highlight the effects of population growth, environmental pollution and energy and resource scarcity, and so herald an emerging era of limits.⁵ Among the solutions proposed in this new era – back-to-the-land, resource conservation and appropriate or alternative technology – was a push for space colonization, led most notably by Princeton physicist Gerard K. O'Neill (1927–92), who in the early 1970s proposed human space settlements based in giant free-floating cylinders (Figure 12.1). In 1974 O'Neill's ideas landed on the front page of the *New York Times*, herald of a widespread movement of technological enthusiasts arguing that humanity could avoid imminent limits by escaping earth.⁶

In 1975, for example, G. Harry Stine (1928–97), an aerospace scientist, model rocket builder and science-fiction author, identified 'a deepening worldwide megacrisis. This megacrisis has many facets. It's an energy crisis. It's a population explosion.' But Stine had good news: 'A far-reaching and final revolution is going to take place in our lifetimes. This revolution has

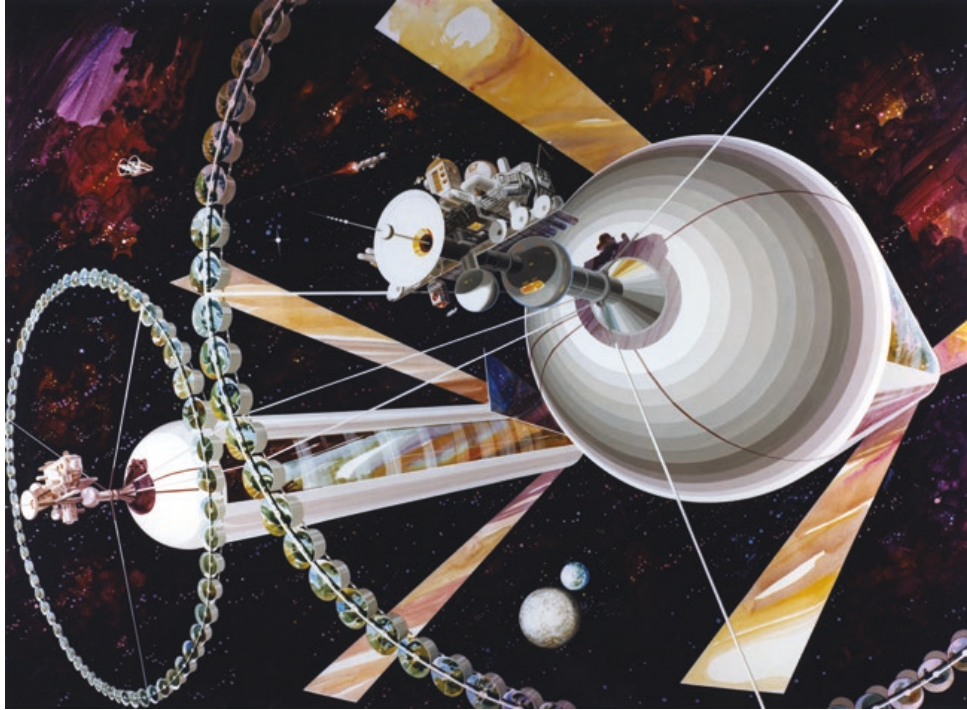


Figure 12.1 Exterior view of a proposed space colony, from studies held at NASA's Ames Research Center in the 1970s to follow up Gerard K. O'Neill's ideas. Artwork by Rick Guidice.

Source: Courtesy of NASA.

already started. [...] It is the Third Industrial Revolution, the final industrial revolution, the exploitation of the Solar System.' Stine's revolution would be driven by the entrepreneur and the capitalist, a stance that resonated especially with libertarians and helped earn the book an introduction by Barry Goldwater (1909–98), a figurehead of conservative politics. Stine, Goldwater enthused, had provided 'a happy combination of hope and optimism, which are vital ingredients of America's greatness. [...] It provides a real challenge to the gloom and doom crowd.'⁷

In 1975, Carolyn (1946–) and Keith Henson (1942–), two devotees of O'Neill's space-colony ideas, formed the L5 Society in Tucson, Arizona. They named it after one of the gravitational equilibrium points in outer space where a space colony could be built. The L5 Society eagerly propagated ideas for space colonies along with literally far-out schemes for lunar and asteroid mining, solar-power satellite systems, solar sails and ion drives. It also provided an early forum for discussion of space-based missile defense. The L5 Society never counted more than 10,000 members, and it might seem like a fringe group, a bunch of *Star Trek* fans off in a corner babbling about ion drives. But it wielded outsized political influence. L5 members, for instance, lobbied to kill the Moon Treaty in 1979, for fear that it would outlaw property rights and hence stall space colonization.⁸

The L5 Society's founders originally responded to the threat to human survival posed by demographic explosion and environmental depletion. Soon, however, the group contemplated that other threat to human survival, the nuclear arms race. The 1970s witnessed an alarming new phase of the arms race, as the Soviet Union matched and then, by 1978, exceeded the earlier build-up of the American arsenal; by that time, each side boasted 25,000 nuclear weapons, including thousands of MIRVed warheads on strategic missiles. Harry Stine's book on the prospects of outer space noted that the current Malthusian crisis was exacerbated by the presence of nuclear weapons, and quoted Krafft Ehricke's warning that 'if we slip back into the Dark Ages this time, we do so with our fingers on the nuclear trigger.'⁹

In May 1976, Keith Henson launched a debate with an article observing that solar-power satellites could run powerful space-based lasers for missile defense. This article is one of the earliest public discussions of such space lasers. Henson at the time was ambivalent about the prospect, questioning the technological, moral and political problems of space-based, beam-weapon missile defenses and noting, 'It is with considerable reluctance that L-5 News opens discussion of this subject.'¹⁰ The article indeed provoked a response from space-colony supporters fearing the militarization of outer space; the editors observed that 'military aspects of space seem to be the Society's most consistent source of controversy.'¹¹ The *L5 News* editors seemed to relish the debate, presenting a series of articles on military space, including excerpts from books downplaying fears of space war and defending military space activity. When one reader suggested that the *L5 News* avoid discussing military uses of outer space, for fear of appearing to support them, the editors replied, 'We'll stop when our under-sand respirators arrive.'¹²

L5's apparent embrace of military space soon came to center on space-based lasers for missile defense and their potential to solve the threat of nuclear war. By June 1979 Keith Henson had abandoned his earlier ambivalence. In an article titled 'Space Forts: Or "Where Are You Obi Wan Kenobi?"', Henson presented a hopeful survey of space-based defenses against the nuclear 'Sword of Damocles' and mocked a recent, skeptical *Scientific American* article on space weapons as having 'the flavor of a 1900 essay on the impossibility of heavier than air flight.' Henson urged military funding of mass drivers and solar sails, which might serve as celestial tugboats to deliver asteroids into earth orbit. Such asteroids could then serve as 'space forts' bristling with lasers or particle beams. A few months later Henson proposed a variant, 'Space Foxholes: Or Beetle Bailey in Orbit,' using material mined from the moon or from Mars's moon Deimos, transported by solar-sail tugs, to protect military satellites.¹³ His wife Carolyn wrote a subsequent article titled 'Lase the Nukes!' that similarly

backed space-based lasers for missile defense as a way to ‘end our fear of nuclear terror.’¹⁴

Keith Henson’s articles suggested, however, that he was perhaps not a true believer in missile defense, but rather viewed it as a means to the end of space colonies. Military programs to deploy large lasers in orbit could develop heavy-lift launch vehicles, space power supplies (such as nuclear reactors or solar-power satellites), mass drivers or solar sails and other expensive space technologies needed for colonizing space. As Henson put it, ‘It’s a long way from military space bunkers to what we are really interested in, the human habitation of space.’ But the military angle justified the investment. ‘The most significant advantage of this project is that it gives us a toehold in space. Once we have a steady supply line for extraterrestrial materials [...] we will have made the step from being visitors in a hostile environment to being homesteaders in the promised land.’¹⁵

Gregory Benford (1941–), a physicist and science-fiction author, including of the popular time-travel novel *Timequake* of 1980, similarly sought to ride military-space coat-tails to space colonies (and similarly invoked frontier rhetoric) in an *Omni* article titled ‘Zeus in Orbit,’ urging the deployment of space-based beam weapons. Benford allowed that military space stations were ‘a far cry from the peaceful scenarios of space industrialization, or from the even less likely vision of farmers in the sky so beloved of the L-5 Society.’ However, he continued, ‘we should face the fact that putting a cap on our nuclear madness is more important.’ The military had the money, and once it had established a toehold in outer space, colonies could follow.¹⁶

Ben Bova (1932–), a former marketing manager for defense firm Avco-Everett (an early producer of high-power lasers) turned *Omni* editor, reversed this approach. In a 1981 book titled *The High Road*, Bova argued, by this time familiarly, that ‘all the disasters we face, from nuclear war to ecological collapse to the tide of irrationality, have one factor in common; population pressure.’ But that assumed that humans were confined to earth. Humanity could tap the vast resources of the solar system (energy, minerals and so on) to relieve pressure on earth – or, better yet, send humans themselves out into space. Bova also noted that both the United States and Soviets were pursuing space-based beam weapons for missile defense. But this was *not* ‘a step away from nuclear holocaust,’ because a defensive shield would encourage the nation that had it to launch a devastating first strike, safe from retaliation. In other words, as SDI critics would soon argue, missile defense was destabilizing. The only way to avoid ‘a new and deadly space race’ was to push the private, commercial development of outer space. Space resources, Bova argued, would ease tensions on earth and civil space systems would crowd out the military in outer space. Military programs need not justify development of civil-space technologies; rather, civilian space programs would prevent the threat of a militarized space.¹⁷

II Strange bedfellows

Missile defense, thanks especially to Reagan and SDI, is often seen as an issue exclusive to conservative politicians. In the 1970s, however, elements of the left-leaning counterculture found themselves, sometimes to their surprise, supporting missile defense alongside more conservative space enthusiasts, in particular libertarians. These 1970s debates over space programs made for strange ideological and cultural bedfellows, from Barry Goldwater to Jerry Brown and Timothy Leary.

One historian of the American space program has stated that by the late 1970s, O'Neill's 'vision to move billions of Earthlings to environmentally correct space colonies looked like a dreamy hangover from the radicalism of the 1960s.'¹⁸ The 1960s influence was not just a dream. LSD guru Timothy Leary (1920–96) supported the L5 Society with his SMI²LE (Space Migration, Intelligence Increase, Life Extension) movement; *Future Life* articles criticized creationism and the Moral Majority, while its letters to the editor included calls for legalizing marijuana. L5's president, Carolyn Henson, in particular imparted an environmentalist, feminist sensibility to the group and borrowed techniques from the anti-war and other protest movements for L5's political activism.¹⁹ Such countercultural currents help explain the vocal response to the military-space issue, with many L5 members lamenting the militarization of outer space.

Carolyn Henson herself wondered how she had gone from protesting against the Vietnam War to supporting space lasers for missile defense. 'Had we repented of our fuzzy-brained leftist credos? Were we victims of the creeping conservatism of middle age? Perhaps we'd traded in our ideals for the promise of more hardware in space? Come now. I'd never admit to imperfection. Let me explain why we veterans of the war against war should back space laser ABM.' Her reasons: space lasers for missile defense were surgical; they were elegant; and, above all, they promised a solution to the threat of nuclear war.²⁰

An exemplar of this cultural overlap appeared in 1978 in a book by J. Peter Vajk, titled *Doomsday Has Been Cancelled*. Vajk's publisher, Peace Press, was founded by activists from Students for a Democratic Society (SDS) in the 1960s to publish anti-war literature, and for the next two decades it provided a consistent home for progressive politics and alternative culture, publishing books by Timothy Leary, marijuana-growing guides, *The Art of Zen Meditation* and *Mysterious Herbs and Roots*. Vajk's book urged space colonization to dispel the era of limits and restore optimism in the human future. Amid discussion of space stations, space mining and space industrialization, Vajk included a section on space-based lasers for missile defense, perhaps powered by solar-power satellites. With such a system 'we may well see Armageddon becoming far less likely by the late 1990s'; humanity could 'escape from the risk of nuclear holocaust.'²¹

Vajk's book mixed references to C.S. Lewis and Charles Lindbergh with ones to Alvin Toffler, Carl Sagan, Fritjof Capra, E.F. Schumacher, Gail Sheehy and Stewart Brand. Vajk's background did not suggest a spaced-out

1960s guru; after getting his PhD in physics from Princeton in 1968, he joined the Theoretical Physics Division at the Livermore weapons lab in California. At Livermore he worked on astrophysics and cosmology, and developed an interest in space colonization, which he then pursued at defense contractor Science Applications International – and as one of the early activists building the L5 Society. The image of a former Livermore physicist smoothly mixing discussion of the Gaia hypothesis and biofeedback with a positive review of space-based missile defense seems jarring only in retrospect.

Vajk was not alone at Livermore in his eclectic interests. L5 board member Lowell Wood, a protégé of Edward Teller, led Livermore's O Group, a freewheeling outfit that developed advanced weapon concepts including the X-ray laser, which would later be the public centerpiece of SDI. Several O-Group members shared Wood's outer-space interests. Peter Hagelstein (1954–) was one of the main developers of the X-ray laser; his dissertation at MIT in 1981, based on his Livermore research, gave a nod to the 'vivid imaginations of the technological dreamers and the wondrous and exciting universes of science-fiction writers,' especially for speculations about beam weapons based on the X-ray laser, and included citations to Larry Niven's *Ringworld*, a popular science-fiction work on space colonies.²²

O Group also included Rod Hyde, who had been inspired by science fiction while in high school, went to MIT to study aeronautical engineering, and then to Livermore as a summer intern in 1972, where he designed a rocket using laser-fusion propulsion. Hyde returned to Livermore while working on his PhD under Wood.²³ A technological optimist, Hyde proved a fertile source of solutions to obstacles on the X-ray laser, especially the optics for overcoming beam diffraction, and hence contributed to Wood's (and Teller's) sanguine projections on its potential for SDI.²⁴ Like Wood, Hyde was an early member of the L5 Society. Later in the 1980s Lowell Wood also hired Stewart Nozette into O Group, after Nozette received his PhD in planetary science from MIT; while in grad school, Nozette had written articles for *L5 News* and *Future Life* about possible mining expeditions in the solar system as a way to dispense with 'gloomy scenarios for our collective futures.'²⁵

L5 fellow-travelers included the original Dr. Strangelove. Herman Kahn (1922–83) earned an image as a cold if not bloodthirsty nuclear strategist for his book *On Thermonuclear War* (1960), a product of his work for the Rand Corporation. He had gone on to help found the Hudson Institute, a futurism think tank, and contributed to a Hudson study backing missile defense.²⁶ In 1976 Kahn and Hudson colleagues published a lengthy report, *The Next 200 Years*, that explicitly challenged pessimistic talk of limits. The study took what Kahn called an 'Earth-centered perspective,' since looking to outer space as a solution seemed like 'a cop-out.' However, Kahn declared, 'It turned out that if you can't solve the problems on Earth, you can solve them in space, as far as we can see.'²⁷

Kahn addressed this comment to California Governor Jerry Brown, noted Democrat, and Amory Lovins, a leading proponent of 'soft' or

‘appropriate’ technology, in a remarkable meeting in Brown’s office in 1976. The conversation between ‘Governor Moonbeam’ and Kahn exemplified the flexible political alignments of this period. Brown had identified the ‘era of limits’ soon after his inauguration in 1975, and had earlier opposed large, centralized technologies as symptomatic of the ills of modern society. But he now embraced outer space with a typically visionary approach. Shortly after this meeting, the *LA Times* commented on the conversion of ‘our new, spaced-out governor’: ‘Gov. Brown is blasting into space. But to achieve lift-off he has had to jettison much of his old rhetorical baggage. He no longer speaks of an “era of limits.” His new high is the “era of possibilities.” Nor is small always beautiful. “In space,” he exults, “big is better.”’²⁸ For his part, Kahn, who later enjoyed a stint as the Barry Goldwater Professor of American Institutions at Arizona State, described himself at the meeting as a neoconservative: ‘Sometimes my friends ask, what do I mean by neoconservative? Well, I’m conservative, but you can’t trust me.’²⁹

Kahn won the admiration of Stewart Brand (1938–), the countercultural, LSD-dropping founder of *Whole Earth Catalog*. Brand published a chapter from *The Next 200 Years* in his *CoEvolution Quarterly*, with an approving editorial comment that urged more consideration of opposing views, singling out the intolerance of space-colony critics.³⁰ He also published the transcript to the Kahn-Brown-Lovins meeting, with more editorial notes expressing his admiration for Kahn: ‘Herman Kahn is to American political intellectuals what Tom Wolfe is to the journalists [...] they are more open, less predictable, and far more informative than your standard ideologue. When it comes to political, social, military, or economic analysis there is no view I’m more interested in than Kahn’s.’ Kahn himself had earlier declared, ‘I like the hippies. I’ve been to Esalen. I’ve had LSD a couple times. In some ways I’d like to join them.’³¹

A year after Kahn’s meeting with Governor Brown and Lovins, he put out a report for NASA with William Brown, a Hudson Institute colleague, on the space program. Like Harry Stine, Kahn and William Brown saw a third watershed in human history underway, after the agricultural revolution of 10,000 BC and the industrial revolution of the nineteenth century. The current shift to post-industrial society (or what they called ‘super-industrial’ society) did not require a space program, but it would surely help: ‘It is almost certain that space exploration will lead to great benefits, and possibly to an extraordinary economic and technological impact’ – and perhaps ‘vast treasures which have not yet been dreamed of.’ Outer space was ‘in any case a true frontier.’ Although the NASA-aimed report mostly skipped over military space, it did mention space-based beam weapons for attacking satellites and also presented, among possible future scenarios, the ‘Great Space War of 2019,’ a one-day conflagration where several spacefaring countries destroyed each other’s satellites and manned space stations. Since the conflict was confined to outer space, the space war was only the ‘moderate’ scenario; the really ‘pessimistic’ scenario was a back-to-nature ‘triumph of

the garden' and rejection of technology. In short, while the report did not see outer space as critical to human future, since humanity could solve its problems with earth-based technologies, it claimed that adding space made things even easier.³²

III Star wars

On 23 March 1983, President Ronald Reagan announced a crash program for what became known as the SDI. SDI would consume some \$26 billion over the next ten years; as a major shift in nuclear strategy from deterrence to defense, SDI was a centerpiece of American military strategy in the late Cold War, a lightning rod for public debate, and a focal point of diplomatic negotiations with the Soviet Union. SDI was often derided as science fiction, as the sort of fantasy technology – death rays, laser guns and space battle stations – common in science-fiction books and films (Figure 12.2). The popular name for the project, 'Star Wars,' derived from the George Lucas film trilogy – released in 1977, 1980 and 1983 – that exemplified the genre. Several writers have suggested the contributions of science fiction to SDI, usually through cultural influence.³³ Some observers, however, perceived a more direct policy influence from science fiction, via the L5 Society. In 1999 science-fiction author Norman Spinrad (1940–) made waves by claiming that his science-fiction colleague Jerry Pournelle (1933–2017) had conceived the crash program for missile defense, sold the Reagan Administration on the idea and even written Reagan's speech. Pournelle replied that he did not write the speech,

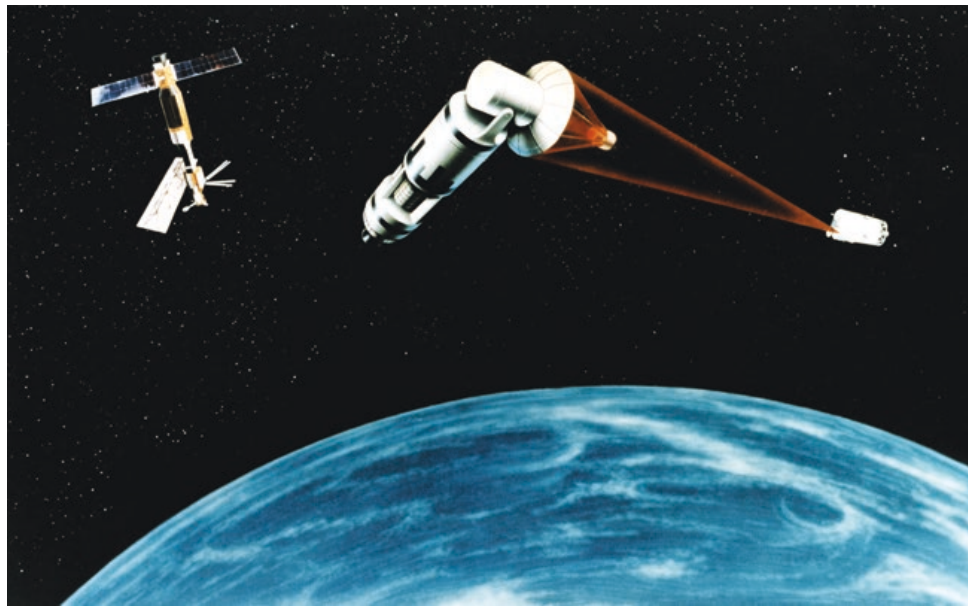


Figure 12.2 A 1984 artist's concept of a space-based laser for SDI.

Source: Courtesy of US Air Force.

but otherwise acknowledged that a group he had led, an offshoot of the L5 Society, was indeed the source of SDI.³⁴

Pournelle was a space-colony enthusiast and an L5 board member in the late 1970s. He had advanced degrees in statistics and systems engineering and PhDs in psychology and political science; he had worked for Boeing and North American-Rockwell before establishing himself as a leading science-fiction author in the 1970s, with tales that combined a strong libertarian bent with technological enthusiasm. His book *A Step Farther Out* of 1979 was an explicit response to the Club of Rome, ‘an effort to show that we are not faced with doom,’ thanks in part to the potential of space colonization.³⁵ Earlier, Pournelle had coauthored, with his mentor Stefan Possony (1913–95), *The Strategy of Technology*, which held that the US competitive advantage in the Cold War lay in high technology; one of its chapters, titled ‘Assured Survival,’ argued for a shift from nuclear deterrence to defense. Pournelle was also one of the science-fiction writers cited as an influence by Peter Hagelstein at Livermore.

In November 1980, Harry Stine was thinking of ways to get the newly elected Ronald Reagan to support an ambitious space-exploration program. Meanwhile, Reagan transition-team staff had asked General Bernard Schriever (1910–2005) to help prepare transition papers on space policy for the incoming Administration. Schriever enlisted a former member of his planning staff, Colonel Francis Kane, and Stine and Kane separately contacted Pournelle, who thought to combine the efforts in a sweeping framework for civilian and military space programs. Pournelle, with support from the L5 Society and the American Astronautical Society, formed a ‘Citizens Advisory Council on National Space Policy.’ The Council convened one weekend at the end of January 1981, soon after Reagan’s inauguration, at the spacious Tarzana home of science-fiction author Larry Niven, Pournelle’s frequent collaborator. The Council counted about 50 members: eight of them from the L5 board; science-fiction authors Robert Heinlein and Poul Anderson; scientists Freeman Dyson, Marvin Minsky, John McCarthy and Eric Drexler; former astronauts Wally Schirra and Gordon Cooper; weapon designers Lowell Wood and Rod Hyde from Livermore; and futurist writers Barbara Marx Hubbard, Harry Stine and Peter Vajk. Thirty members attended this first meeting, with Pournelle as chairman.³⁶

These initial meetings led to a Council report, in spring 1981, titled ‘Space: The Crucial Frontier.’ Its second and third sentences declared: ‘Progress is possible. We do not have to accept limits to growth.’ Among chapters recommending space stations, space solar power satellites, space commerce and a manned lunar colony by 2000, the report included a chapter urging space-based laser defenses.³⁷ According to Pournelle, the Council had a direct line to Reagan’s National Security Adviser, Richard V. Allen (1936–), a protégé of Council member Possony; Allen got the Council’s report from Possony and brought its message to Reagan.³⁸ Heinlein supported this claim

in a 1985 letter to Barney Oliver, SETI scientist and SDI skeptic, that also expressed hardening ideological positions: ‘It was endless effort by a mere handful of us that got the matter to Mr. Reagan’s attention and resulted in his “Star Wars” speech.’ Heinlein viewed the issue in the starkest light: ‘Please understand that this is not just a friendly debate, this is not the world series; this is grim death, the very survival of our country.’ He concluded with a query to Oliver: ‘Which side are you on, Doctor? “Better Red than Dead?” Or “Better Dead than Red?”’³⁹

Spinrad’s 1999 article reintroduced the old coat-tails argument, attributing to Pournelle the strategy of supporting SDI as a clever way to attain a long-range space colony.⁴⁰ If so, that would add an interesting twist to SDI. Some Reagan advisers have claimed that SDI was a clever sting operation, to force the Soviets to spend themselves into bankruptcy building high-tech space systems.⁴¹ If Spinrad was right, SDI was indeed a sting operation – against America’s own military, convincing it to spend money on a chimerical military program that would eventually help humanity populate outer space. Pournelle, however, was a true believer and a strong anti-communist, evident in his 1970 work with Possony on strategic defense and his authoring of the Council report chapter on space-based missile defense; he also wrote a popular article for *Omni* in 1981 warning about Soviet efforts in space-based beam weapons and urging the United States to prepare for space war.⁴²

These suggestive links do not mean that the Council created Reagan’s policy. There is no contemporary evidence that they played a direct policy role in the Reagan Administration’s shift from strategic deterrence to defense.⁴³ The actual seeds for SDI came rather from technological developments including directed-energy weapons, such as the X-ray laser, but far more so in micro-electronics, phased-array radars and infrared focal-plane arrays; from strategic debates over MX missile deployment and ballistic-missile submarine vulnerability; and moral and political concerns about the nuclear arms race, including the Freeze movement and the Catholic bishops’ letter on nuclear weapons.

The L5 Society and space enthusiasts in general, however, were among the groups that prepared the ground for SDI through public arguments for space-based missile defense. And the L5 Society was one of the first, starting with Keith Henson’s article of 1976, and prominent L5 members became important SDI advocates, in particular Lowell Wood, who made the X-ray laser a centerpiece of the public debate over SDI. The L5 Society also provided more concrete connections thanks to institutional links between L5 and the High Frontier organization, created by General Daniel Graham (1925–95) in 1981 as a political lobby for missile defense. Graham hired several administrative staff members from the L5 Society, including its secretary, who brought with them the Society’s political-action experience and, more important, its mailing lists. Pournelle later claimed that ‘Dan Graham essentially took the remnants of the L5 Society and built his High Frontier out of it.’⁴⁴

The High Frontier group supplemented the grassroots, L5-style public approach with kitchen-cabinet lobbying by Reagan's friends, such as Karl Bendetsen, Jaquelin Hume and Joseph Coors, that helped bring missile defense more to the Administration's attention.⁴⁵ Early L5 activists Pournelle and Vajk also contributed to Graham's *High Frontier* report, for which Heinlein wrote the foreword. The report echoed not only the title of Gerard K. O'Neill's original opus on space colonies, but also the original name of what became the L5 Society. Another contributor to Graham's report, John Bonsma, soon afterward married L5 founder Carolyn Meinel, who had divorced Keith Henson. Meanwhile, Pournelle's next book, *Mutual Assured Survival*, written with Dean Ing and based on the Council's report, boasted a blurb from Reagan.⁴⁶

The space-colony and science-fiction influence appears also in the rhetoric deployed in Reagan's announcement of SDI. One of the most striking lines in his March 1983 speech called on the scientific community, the creators of nuclear weapons, 'to give us the means of rendering these weapons impotent and obsolete.' In 1978 Vajk's book on space colonization had declared that space-based lasers 'would render ICBMs totally obsolete,'⁴⁷ and Benford's 1981 *Omni* article, 'Zeus in Orbit,' carried the subtitle, 'Particle-beam weapons threaten to scuttle ballistic missiles and make the MX obsolete.'⁴⁸

IV Transcending limits

SDI usually appears strictly as a product of the Cold War. This chapter broadens the context for SDI to consider not only the strategic developments of the late 1970s and early 1980s, but also the crisis earlier in the 1970s sparked by *Limits to Growth* and other studies. But this chapter also shows that the era of limits included the threat of nuclear Armageddon, and even in utopian space forums such as the L5 Society the Cold War crept in.

The L5 Society connected the Club of Rome to SDI and the romantic counterculture to military space programs. These connections forced some L5 members to overcome ambivalence regarding the militarization of space, but they also provided SDI with one of its key characteristics. L5 was among the first groups to publicly argue for space-based missile defense, but that was not all. L5 injected its utopianism into the missile defense issue, and thus helped change ideas about missile defense itself. L5's space cadets urged a shift in missile defense from the existing ground-based, incremental approach to a vastly more ambitious, space-based, beam-weapon, Buck-Rogers program. Instead of methodically climbing the ladder from familiar ground-based rocket interceptors, the L5 crowd wanted to shoot straight into space, at warp speed and with phasers set to kill. That is indeed what Reagan proposed with SDI, and this breathtaking ambition was what set SDI apart. That ambition was spurred by groups like L5 in the 1970s, who saw outer space as a way to make strategic weapons 'obsolete.' And this ambition, in turn, was energized by the belief, popularized by the Club of Rome, that there was an existential crisis facing humankind.

SDI and missile defense continued to shape nominally civilian space programs, from reusable launch vehicles to planetary spacecraft to infrared astronomy; Lowell Wood and Rod Hyde kept alive the space-colony dream in the early 1990s with proposals for inflatable space stations using SDI concepts.⁴⁹ And military space interests continued to engage the L5 Society; the society itself fractured in the mid-1980s along political lines, thanks in part to SDI, and L5 eventually merged with the National Space Society, which was more closely aligned with the aerospace industry.⁵⁰ But space-colony supporters continued to argue that space-weapon programs ‘can ensure the growth of infrastructure and enable the establishment of human settlements in space.’ Odd political alignments also persisted, though more recently on the other side of the issue: whereas in the 1970s liberals and libertarians could agree on the merits of space-based missile defense, in the 2000s a proposed ban on space weapons was supported from the left (the Berkeley city council) and the far right (militia movements challenging space-based surveillance systems as threats to individual liberty).⁵¹

The L5 story sheds light on the context of the 1970s, when far-out ideas – asteroid mining, biofeedback or space-based missile defense – could receive a hearing at either end of the ideological spectrum. It also demonstrates how the far left and far right could, like a closed universe, bend around and meet at particular points.⁵² In 1983, when L5 had already grown more conservative, an unpublished survey of its members found 31 percent identified themselves as Republicans, 25 percent as Democrats, 8 percent as Libertarians and 33 percent as none of the above. Addressing an L5 conference the next year, Congressman George E. Brown (1920–99) declared rather that L5 members were ‘5 percent Democrat, 5 percent Republican, and 90 percent anarchist.’⁵³

This episode occurred amid an evolving ideological landscape for the general American polity and for the space program in particular. Andrew Butrica has detailed the emergence of a conservative space agenda in the 1980s under Reagan, but its roots reach deeper than Reagan’s election.⁵⁴ The 1970s marked a conservative ideological turn in the United States, as the New Right merged evangelical moralists, Goldwaterite populists, free-market economists and neoconservatives.⁵⁵ Conservatives, traditionally skeptical of the Enlightenment faith in progress and improbability of humankind, became technological enthusiasts. The period also coincided with an ideological sea change for the US space program. As the highest expression of socially directed technical innovation, the early space program received its main support from politicians on the left. But political liberals in the 1960s began directing federal spending toward social problems instead of technoscientific extravaganzas that seemed only to benefit a few scientists and aerospace corporations. As liberals drifted down to earth, conservatives were abandoning fiscal austerity and their traditional aversion to Enlightenment faith in human progress. Conservatives became technological enthusiasts and embraced the vision of space as new frontier: the space program could rekindle the old pioneer spirit,

inspiring noble achievements and opening up a new realm for commerce. Some liberal commentators for their part came to view the frontier myth as an emblem of imperial conquest, environmental damage, selective government subsidies and corporate profiteering.⁵⁶

The 1970s provided a moment of flux before new alignments had solidified, when alternative connections could be explored. Thus, for example, despite the countercultural past of some of the L5 Society's leaders, the group embraced the rhetoric of colonization. Outer space was a frontier not just to explore, but also to settle and exploit; hence their references to 'farmers in the sky' and 'homesteaders in the promised land.' The latter echoed the eschatological imagery of the Club of Rome itself, which to some observers seemed to claim that the world was going to hell, figuratively if not literally, and the solution lay in more virtuous living. That was the view of its critics: Kahn viewed the Club of Rome as 'an exercise in moralistic judgments,' a 'package of New Testament virtues and preindustrial simplicity,' with whiffs of 'hellfire and sin.'⁵⁷ L5 critics in turn detected a revivalist, religious bent to the space-colony movement, with O'Neill as 'prophet.'⁵⁸ One *L5 News* reader wrote in protest of a 'pseudo-evangelical/fundamental editorial viewpoint,' claiming that "'Space Salesmen" seem very much the harbingers of a new religion. A religion promising to wipe away the problems of war and peace, of population, of nuclear war and wastes, of diminishing resources.'⁵⁹ Space colonists, however, countered the virtuous moralism of the 'era of limits' with faith in science and technology. When L5 enthusiasts talked about rising up to reside in the heavens, they meant it literally.

The quasi-spiritual dimension to these debates perhaps reflects the countercultural influence, which injected the romantic, utopian idealism of the 1960s with technological enthusiasm. How could former 1960s feminists, environmentalists and anti-war protesters support missile defense? The 1960s counterculture is supposed to represent a backlash against science and technology, which had become associated with the military, environmental pollution and technocratic challenges to democracy. Technology was the source of society's problems, not the solution. But science and technology were not necessarily opposed to the counterculture. Science and technology reflect their social context, and for the 1960s and 1970s that included the counterculture, as historians are beginning to recognize.⁶⁰

This combination of cosmic idealism and technological enthusiasm was not new, nor confined to the United States. Consider the ostensible target of missile defense, the Soviet Union, whose active space program at the time aroused envy among some American space enthusiasts. The *L5 News* provided regular updates on Soviet efforts, reprinting articles on Soviet programs and noting that 'scientific bigthink is a fully approved literary genre in the Soviet Union.'⁶¹ In 1981, American space writer James E. Oberg (1944–) predicted that by 2000 the Soviets would have dozens of 'space outposts' with hundreds of residents, 'the first true "space colonists".'⁶² Soviet interest likewise had deep

utopian roots. Although era-of-limits rhetoric certainly did not jibe with Soviet ideology, Russian and Soviet space enthusiasts had long engaged Malthusian concerns. In the late nineteenth century Nikolay Fyodorov (1829–1903) urged space colonization to relieve population pressure. Fyodorov's disciple Konstantin Tsiolkovsky (1857–1935), who provided the technical and popular foundations of Soviet astronautics and popularized the notion of spaceflight, helped form the Anarcho-Biocosmists (or Cosmists, for short) in the 1920s.⁶³ Asif Siddiqi has shown how the Cosmists, along with science-fiction authors such as Aleksey Tolstoy (1882–1945) inspired several amateur space societies in the Soviet Union in the 1920s and 1930s that combined mystical and spiritual idealism with technological enthusiasm.⁶⁴ To the Cosmists, faith and reason were not mutually exclusive; utopian mysticism and romantic idealism resonated with science and technology.

The space-colony advocates in the 1970s US similarly did not reject science or push for decentralized, small-scale, 'appropriate' technology; on the contrary, they openly embraced big, high-tech systems in their idealistic quest to save the planet. The common denominator of the L5 Society, Gerard K. O'Neill, Herman Kahn, Livermore's O Group, Stewart Brand and Jerry Brown was technological enthusiasm. They quelled the doom-and-gloom talk of the 'era of limits' with their belief that technology could solve existential threats, including that posed by nuclear weapons. The 1970s space cadets helped revive faith in high technology – in this case, very high technology.

Notes

1. California governor Jerry Brown popularized the 'era of limits' term in campaign speeches in 1976. See, for example, William Endicott, 'Brown, on TV, Talks Issues, Concedes Carter is Strong but Retains Delegates,' *Los Angeles Times* (26 June 1976), A5, and Orville Schell, 'Jerry Brown: Economics in an Era of Limits,' *Los Angeles Times* (5 August 1979), G1; see also Dominic Sandbrook, *Mad as Hell: The Crisis of the 1970s and the Rise of the Populist Right*, New York: Anchor Books, 2012, 275–6. The Club of Rome's original report was Donella H. Meadows, Dennis L. Meadows, Jørgen Randers, William W. Behrens III and Club of Rome, *The Limits to Growth: A Report for the Club of Rome's Project on the Predicament of Mankind*, New York: Universe Books, 1972. I thank Patrick McCray for many conversations and generous sharing of sources. Archived issues of *L5 News* are available at www.nss.org/settlement/L5news (accessed 1 October 2017).
2. Francis FitzGerald, *Way Out There in the Blue: Reagan, Star Wars, and the End of the Cold War*, New York: Touchstone, 2000; Paul Lettow, *Ronald Reagan and his Quest to Abolish Nuclear Weapons*, New York: Random House, 2005; Nigel Hey, *The Star Wars Enigma: Behind the Scenes of the Cold War Race for Missile Defense*, Washington, DC: Potomac Books, 2006. Histories of SDI based more heavily on archival research are Donald R. Baucom, *The Origins of SDI, 1944–1983*, Lawrence: University Press of Kansas, 1992, and Andrew Butrica, *Single Stage to Orbit: Politics, Space Technology, and the Quest for Reusable Rocketry*, Baltimore: Johns Hopkins University Press, 2003.

3. Peter J. Westwick, "Space-Strike Weapons" and the Soviet Response to SDI,' *Diplomatic History* 32.5 (November 2008), 955–79.
4. Idem, 'The International History of the Strategic Defense Initiative: American Influence and Economic Competition in the Late Cold War,' *Centaurus* 52.4 (Fall 2010), 338–51.
5. Paul R. Ehrlich, *The Population Bomb*, New York: Ballantine Books, 1968; Roberto Vacca, *The Coming Dark Age*, Garden City: Doubleday, 1973.
6. On O'Neill, see 'Future Vacation Spot: The L5 Libration Point,' *New York Times* (19 May 1974), 6; and W. Patrick McCray, *The Visioneers: How a Group of Elite Scientists Pursued Space Colonies, Nanotechnologies, and a Limitless Future*, Princeton: Princeton University Press, 2012. On appropriate technology, see Ernst Friedrich Schumacher, *Small is Beautiful: Economics as if People Mattered*, New York: Harper Perennial, 1989, 190–201; and Carroll Pursell, 'The Rise and Fall of the Appropriate Technology Movement in the United States, 1965–1985,' *Technology and Culture* 34.3 (July 1993), 629–37. See also Amory B. Lovins, *Soft Energy Paths: Toward a Durable Peace*, San Francisco: HarperCollins, 1977.
7. G. Harry Stine, *The Third Industrial Revolution*, New York: Putnam, 1975, 14, 8.
8. On the Moon Treaty and other examples of L5 activism, see Michael A.G. Michaud, *Reaching for the High Frontier: The American Pro-Space Movement, 1972–84*, New York: Praeger, 1986, esp. 90–4, and Luca Follis's contribution, Chapter 8 in this volume.
9. Stine, *Third Industrial Revolution*, 14; see also Robert Poole's contribution, Chapter 5 in this volume.
10. Keith Henson, 'Military Aspects of SSPS Power,' *L5 News* 1.9 (May 1976), 2.
11. John Holt, 'Space Research and the Military,' *L5 News* 2.2 (February 1977), 3–4; 'Space Warfare Replies' [quote], *L5 News* 2.7 (July 1977), 6–7; see also letters to editor, *L5 News* 2.4 (April 1977), 14–15.
12. Michael Mautner to editor, *L5 News* 1.11 (July 1976), 5, and editor's response; James E. Oberg, 'Facts and Fallacies of Space Warfare,' *L5 News* 3.10 (October 1978), 9–11, and idem, 'Space Wars,' *L5 News* 4.4 (April 1979), 10–11.
13. Keith Henson, 'Space Forts: Or "Where Are You Obi Wan Kenobi?"', *L5 News* 4.6 (June 1979), 1–3; Keith Henson, 'Space Foxholes: Or Beetle Bailey in Orbit,' *L5 News*, 4.10 (October 79), 3–4.
14. Carolyn Henson, 'Lase the Nukes!,' *Future Life* 26 (May 1981), 21, 42.
15. Henson, 'Space Foxholes.'
16. Gregory Benford, 'Zeus in Orbit,' *Omni* 3.12 (September 1981), 53–5, 115–16.
17. Ben Bova, *The High Road*, Boston: Houghton Mifflin, 1981, 13, 206–18.
18. Howard E. McCurdy, *Space and the American Imagination*, Washington, DC: Smithsonian Institution Press, 1997, 154.
19. Harlan Ellison, 'An Edge in My Voice,' *Future Life* 27 (June 1981), 54; on Leary and on Carolyn Henson, see Michaud, *Reaching for the High Frontier*, 86–9.
20. Henson, 'Lase the Nukes!,' 21, 42. Carolyn's father, Aden Meinel, had followed a similar path from anti-war protester to missile-defense supporter.
21. J. Peter Vajk, *Doomsday Has Been Cancelled*, Culver City: Peace Press, 1978, 160.
22. Peter L. Hagelstein, *Physics of Short Wavelength Laser Design*, PhD thesis, Lawrence Livermore National Laboratory, 1981.
23. William J. Broad, *Star Warriors: A Penetrating Look into the Lives of the Young Scientists Behind Our Space Age Weaponry*, New York: Touchstone, 1985, 119–20, 131.

24. George Chapline interview, 19 April 2007; Robin Staffin interview, 6 November 2007, both with the author.
25. Stewart Nozette quote from 'Prospecting in the Solar System,' *Future Life* 21 (September 1980), 40–3; idem, 'Intrinsically Valuable Materials in Space,' *L5 News* 3.6 (June 1978), 5–6; idem, 'The Asteroid/Meteorite Connection,' *L5 News* 4.10 (October 1979), 6–7. Nozette recently made headlines on charges of attempted espionage for Israel; see Peter Grier, 'Stewart Nozette: The American Who Wanted to Spy for Israel,' *Christian Science Monitor* (21 October 2009), 2.
26. Johan J. Holst and William Schneider Jr., eds, *Why ABM?*, New York: Pergamon Press, 1969.
27. Herman Kahn, Jerry Brown and Amory Lovins, 'The New Class,' *CoEvolution Quarterly* 13 (Spring 1977), 8–39.
28. 'Our New, Spaced-out Governor,' *LA Times* (5 August 1977), D6. Joseph Lelyveld, 'Jerry Brown's Space Program,' *New York Times Magazine* (17 July 1977), 182. In a 1982 interview Newt Gingrich (1943–) noted his curious alignment with Brown, but suggested that 'both Jerry Brown and I are trying to find a way to re-create the spirit of optimism and vision of hope that once characterized America'; see Miles Beller, 'Q&A with Gingrich,' *LA Herald Examiner* (13 April 1982), 2, 4.
29. Kahn et al., 'The New Class.'
30. Herman Kahn, William Brown and Leon Martel, 'From Present to Future: The Problems of Transition to a Postindustrial Society,' *CoEvolution Quarterly* 11 (Fall 1976), 4–17.
31. Kahn et al., 'The New Class,' 8 [Brand], 10, and 24 [Kahn]; Kahn quote on hippies from 1968, in Fred Turner, *From Counterculture to Cyberculture: Stewart Brand, the Whole Earth Network, and the Rise of Digital Utopianism*, Chicago: University of Chicago Press, 2006, 186. Kahn also said, 'I stylize myself as a Stoic. My wife says, "You can't be a 300-pound Stoic," so I'm a neo-Stoic.' Quoted in Kahn et al., 'The New Class,' 36. On Kahn, see also Sharon Ghamari-Tabrizi, *The Worlds of Herman Kahn: The Intuitive Science of Nuclear War*, Cambridge, MA: Harvard University Press, 2005.
32. William M. Brown and Herman Kahn, 'Long-Term Prospects for Developments in Space (A Scenario Approach),' Hudson Institute report HI-2638-RR, NASA report CR-156837, Croton-on-Hudson: Hudson Institute, 1977, 77, 218, 223–4, 259.
33. On science-fiction links to SDI, see William J. Broad, 'Science Fiction Authors Choose Sides in "Star Wars,"' *New York Times* (26 February 1985), C1. Thomas M. Disch, 'The Road to Heaven: Science Fiction and the Militarization of Space,' *The Nation* (10 May 1986), 650–6; H. Bruce Franklin, *War Stars: The Super-weapon and the American Imagination*, Oxford: Oxford University Press, 1988, 199–201; Chris Hables Gray, '"There Will Be War!": Future War Fantasies and Militaristic Science Fiction in the 1980s,' *Science Fiction Studies* 21.3 (1994), 315–36; David Seed, *American Science Fiction and the Cold War*, Edinburgh: Edinburgh University Press, 1999, 181–93. Michael P. Rogin, *Ronald Reagan, The Movie: And Other Exercises in Political Demonology*, Berkeley: University of California, 1987, was one of the first to argue that SDI was a case of life imitating art, through Reagan's own acting career, especially *Murder in the Air*, the 1940 film that featured an 'inertia projector' similar to SDI-era beam weapons. Rebecca

- Slayton has examined how popular images of lasers shaped their role in SDI; see idem, 'From Death Rays to Light Sabers: Making Lasers Surgically Precise,' *Technology and Culture* 52.1 (January 2011), 45–74.
34. Norman Spinrad, 'Too High the Moon: From Jules Verne to Star Wars,' *Le Monde Diplomatique* (July 1999), posted at mondediplo.com/1999/07/14star (accessed 1 October 2017); Jerry Pournelle, 'Chaos Manor debates,' 25 February 2007; available at www.jerrypournelle.com/debates/nasa-sdi.html#respond1 (accessed 1 October 2017).
 35. Idem, *A Step Farther Out*, New York: Ace Books, 1979, 3.
 36. The *L5 News* from April 1981 has photos and a summary of the meeting. See also 'Greg Bear Interview Part 2: The Bullet You Don't Hear,' *PBS Wired Science interview* (22 December 2007); available at <http://www.pbs.org/kcet/wiredscience/blogs/2007/12/> (accessed 28 May 2014).
 37. Citizens Advisory Council on National Space Policy, 'Space: The Crucial Frontier,' Spring 1981; available at www.nss.org/settlement/L5news/1981-council.htm (accessed 1 October 2017).
 38. Pournelle interview by Patrick McCray and Peter Westwick, 12 October 2007.
 39. Heinlein to Barney Oliver, 22 February 1985, Robert Heinlein papers, UC Santa Cruz.
 40. Spinrad, 'Too High the Moon.'
 41. For a short summary of the sting theory, see Hendrik Hertzberg, 'Laser Show,' *New Yorker* 76.11 (15 May 2000), 92–6. The theory is presented, for instance, in Robert C. McFarlane, *Special Trust*, New York: Cadell and Davies, 1994, and Peter Schweizer, *Victory: The Reagan Administration's Secret Strategy that Hastened the Collapse of the Soviet Union*, New York: Atlantic Monthly Press, 1994.
 42. Jerry Pournelle, 'Space: The Decisive Frontier,' *Omni* 4.2 (November 1981), 30, 138–9.
 43. One could also argue that SDI did not produce a major shift in policy, and that deterrence continued as the underlying principle of US nuclear doctrine: see Janne E. Nolan, *Guardians of the Arsenal: The Politics of Nuclear Strategy*, New York: Basic Books, 1989.
 44. Pournelle interview; on L5's political activity, see Michaud, *Reaching for the High Frontier*, 85–98.
 45. Bendetsen to Joseph Coors, 25 October 1982; Coors to Bendetsen, 18 October 1982; Weinberger to Graham, 24 November 1982; Clark to Graham, 6 December 1982; Teller to Bendetsen, 29 December 1982; Bendetsen to Keyworth, 27 December 1982; and Bendetsen et al., 'Proposal for Inclusion in the President's State of the Union Address,' all in Edward Teller papers, box 273/folder 'Bendetsen,' Hoover Institution, Stanford University. Joseph Coors interview by Donald Baucom, 31 July 1987, and Jaquelin Hume interview by Donald Baucom, 29 October 1987, SDI Interview Collection, Ronald Reagan Presidential Library, Simi Valley, CA.
 46. Michaud, *Reaching for the High Frontier*, 228–9. Jerry Pournelle and Dean Ing, *Mutual Assured Survival*, New York: Baen Enterprises, 1984.
 47. Vajk, *Doomsday*, 160. Vajk may have picked it up from the Hitchcock film *Torn Curtain* of 1966, in which an American agent played by Paul Newman says the line, 'We will produce a defensive weapon that will make all nuclear weapons obsolete.' FitzGerald, *Way Out There in the Blue*, 23.

48. Gregory Benford, 'Zeus in Orbit,' *Omni* 3.12 (September 1981), 53–5, 115–16. Reagan first used the term after the Joint Chiefs meeting in February 1983 that reconsidered missile defense, described by Reagan in his diary: 'Out of it came a super idea [...] a defensive weapon that could make nuclear weapons obsolete.' Ronald Reagan, *The Reagan Diaries*, ed. Douglas Brinkley, New York: HarperCollins, 2007, s.v. 11 February 1983. Pournelle has stated another famous line from Reagan's speech, 'wouldn't it be better to save lives than to avenge them,' was suggested by Jim Baen, editor of the science-fiction magazine *Galaxy*, at the Citizens Council meeting in 1981. The quote does not appear in the Council's report.
49. Butrica, *Single Stage*; Peter J. Westwick, *Into the Black: JPL and the American Space Program, 1976–2004*, New Haven: Yale University Press, 2007. See also Felicity Mellor, 'Colliding Worlds: Asteroid Research and the Legitimization of War in Space,' *Social Studies of Science* 37.4 (August 2007), 499–531.
50. Michaud, *Reaching for the High Frontier*, 243–4; Pournelle interview by McCray and Westwick.
51. John Carter McKnight, 'Let's Weaponize Space,' *Space Daily* (30 January 2003); available at www.spacedaily.com/news/oped-03d.html (accessed 1 October 2017).
52. John Judis, 'Libertarianism: Where the Left Meets the Right,' *Progressive* 44 (September 1980), 36–8.
53. Michaud, *Reaching for the High Frontier*, 120, 97.
54. Butrica, *Single Stage*.
55. Laura Kalman, *Right Star Rising: A New Politics, 1974–1980*, New York: Norton, 2010; Sandbrook, *Mad as Hell*; on the 1970s in general, see also Niall Ferguson, Charles S. Maier, Erez Manela and Daniel J. Sargent, eds, *The Shock of the Global: The 1970s in Perspective*, Cambridge, MA: Harvard University Press, 2010. The extensive literature on neocons includes Jacob Heilbrunn, *They Knew They Were Right: The Rise of the Neocons*, New York: Doubleday, 2008; James Mann, *Rise of the Vulcans: The History of Bush's War Cabinet*, New York: Penguin, 2004; Murray Friedman, *The Neoconservative Revolution: Jewish Intellectuals and the Shaping of Public Policy*, Cambridge: Cambridge University Press, 2005; and Anne Norton, *Leo Strauss and the Politics of American Empire*, New Haven: Yale University Press, 2005. Some commentators argue that no right turn occurred; see, for example, Thomas Ferguson and Joel Rogers, 'The Myth of America's Turn to the Right,' *Atlantic Monthly* 257.5 (May 1986), 43–53.
56. Roger D. Launius and Howard E. McCurdy, 'Epilogue: Beyond NASA Exceptionalism,' in eadem, eds, *Spaceflight and the Myth of Presidential Leadership*, Urbana: University of Illinois Press, 1997, 221–50, here 234–40.
57. James Reston Jr., 'The Wrath of Kahn,' *Omni* 5.12 (September 1983), 69–76. Literal biblical millennialism appeared at the time in the 1970 bestseller, *The Late Great Planet Earth*, and Reagan himself embraced end-times eschatology; see Richard N. Ostling, Michael P. Harris and James Castelli, 'Armageddon and the End Times,' *Time* 124.19 (5 November 1984), 73. Frances FitzGerald has argued that Reagan used SDI as an expression of America's civic religion, following G. Simon Harak's earlier identification of the 'soteriology of SDI.' See FitzGerald, *Way Out There in the Blue*; and Harak, 'One Nation, under God: The Soteriology of SDI,' *Journal of the American Academy of Religion* 56.3 (1988), 497–527.

58. Michaud, *Reaching for the High Frontier*, 98.
59. Gary Goodman, letter to editor, *L5 News* 2.7 (July 1977), 6–7.
60. David Kaiser, *How the Hippies Saved Physics: Science, Counterculture, and the Quantum Revival*, New York: Norton, 2012; Michael D. Gordin, *The Pseudoscience Wars: Immanuel Velikovsky and the Birth of the Modern Fringe*, Chicago: University of Chicago Press, 2012.
61. Iosif Zorich, 'Energy from Space,' reprinted in *L5 News* 4.6 (June 1979), 5–6; Dick Fredericksen, 'When the Soviets Let Their Hair Down,' *ibid.*, 4.
62. James E. Oberg, *Red Star in Orbit*, New York: Random House, 1981, 235–6. Soviet writers criticized American references to 'space colonization' as another example of American imperialism. Frank H. Tucker, 'Soviet Science Fiction: Recent Developments and Outlook,' *Russian Review* 33.2 (1974), 189–200, here 199.
63. Paul Avrich, 'Russian Anarchists and the Civil War,' *Russian Review* 27.3 (1968), 296–306; James T. Andrews, *Science for the Masses: The Bolshevik State, Public Science, and the Popular Imagination in Soviet Russia, 1917–1934*, College Station: Texas A&M University Press, 2003, 85–6; Daniel Gerould, 'On Soviet Science Fiction,' *Science Fiction Studies* 10.3 (November 1983), 341–3. Fyodorov and the Cosmists also speculated that space travel could lead to immortality, another idea revived in the Soviet Union in the 1970s – at the same time as an American revival; see McCray, *Visioneers*.
64. Asif A. Siddiqi, 'Imagining the Cosmos: Utopians, Mystics, and the Popular Culture of Spaceflight in Revolutionary Russia,' *Osiris* 23.1 (2008), 260–88; Aleksey Tolstoy, *The Garin Death Ray*, Moscow: Foreign Languages Publishing House, 1955. Tolstoy entranced generations of Soviet scientists and engineers with his tales of space travel and, in his science-fiction novel *The Garin Death Ray* of 1927, laser-like beam weapons; Peter Zarubin interview by Peter Westwick, 5 October 2006.